

Main objective of this Chabot:

To answer questions about HR policies, benefits, payroll, leave management, etc. Service available 24/7

Key features to be considered:

Natural Language Understanding (NLU): For accurately interpreting employee queries

Integration with HR Systems: Connect to HRMS, payroll, and collaboration tools (e.g., Slack, Teams)

Channel: Web and Mobile

Personalization: Remember user preferences and previous interactions for tailored responses

Live Agent Handoff: Seamless escalation to human HR staff when needed

Analytics and Reporting: Track usage, common queries, and employee sentiment for HR insights

Multilingual Support: Let us try English and Tamil

Continuous Learning: Improve over time using feedback and new HR data

Tech stack to be used:

Programming language : Python

Framework : Rasa or Botpress

UI (For testing): Streamlit

UI (For Production): React JS

Database: MySQL or Maria DB

Action Items:

1. Gather and Prepare Data

Collect HR policies, FAQs, historical queries, and integrate with HR systems for real-time data

2. Design Conversation Flows

Map out user journeys for each HR task; ensure clarity and a friendly, professional tone

3. Choose Platform/Framework

Select based on integration needs, scalability, and team expertise

4. Develop and Train the Chatbot

Use NLU to handle varied queries; implement RAG (Retrieval-Augmented Generation) for accurate, context-aware responses

5. Integrate with HR Systems

Connect to HRMS, payroll, and communication platforms for seamless workflow

6. Test and Iterate

Conduct user testing, gather feedback, and refine conversation flows and responses

7. Deploy and Monitor

Roll out to employees, monitor usage, and update content as policies and processes evolve

8. Continuous Improvement

Use analytics and feedback to enhance Chabot capabilities and user experience

Summary
Table

Action Item	Key Activities	Tools/Tech
Gather Data	Collect, clean, tag, and structure HR info	Spreadsheets, Python
Design Flows	Map user journeys, create conversation diagrams	Draw.io, Lucidchart
Choose Framework	Evaluate Rasa/Botpress, check integration support	Rasa, Botpress
Develop & Train	NLU, RAG, multilingual, personalization	Python, Rasa/Botpress
Integrate Systems	Connect APIs, ensure security, sync data	REST/GraphQL, Python
Test & Iterate	User testing, feedback, flow refinement	Streamlit, logs
Deploy & Monitor	Launch UI, set up analytics and monitoring	React, ELK, Streamlit
Continuous Improve	Retrain, update content, analyze, engage users	Python, dashboards